

Vibe Shuffle: Next-Song Selection from Relative Valence–Arousal Trajectories

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Abstract

Music recommenders learn what listeners like in general, but the song that fits depends on the moment. Vibe Shuffle asks whether within-session camera and cardiac observations can make next-song selection more context-sensitive. At the start of a session, the system initializes personal references for facial and cardiac signals. Later facial and cardiac observations are expressed as changes from those references and combined with movement and nodding summarized over a recent camera window. Together, these observations form a short trajectory in a valence–arousal plane without assigning the listener an emotion label. The trajectory mean identifies a catalog region, and the nearest unplayed song in that region is selected. An exploratory within-session crossover compared Vibe Shuffle with Random selection from the same catalog. Mood-fit ratings were higher on average under Vibe Shuffle while general liking differed little on average, but both confidence intervals included zero, so no recommendation benefit is established. The contribution is an inspectable method that turns within-session multimodal observations into a single song-selection decision, together with a blueprint for the study needed to test it.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Information systems** → *Recommender systems*.

Keywords

affective computing, music recommendation, relative affect, valence and arousal, heart-rate variability, facial cues, personal references

1 Introduction

Music recommendation is usually based on what a person tends to like. Listening history, favorite artists, similar users, and song features all provide useful signals of long-term preference. Yet the song that fits can change from one moment to the next. A track that works during exercise may feel out of place during a quiet break, and a song that someone likes in general may still not be the song they want to hear next. Because people use music to maintain, change, and regulate mood and activation [12, 26], this difference between general liking and momentary fit is an important challenge for contextual music recommendation [13, 27].

One way to include the current situation is to ask listeners to select their mood. This can provide useful context, but it also interrupts the listening experience. Automatic adaptation avoids that interruption, although the available observations are uncertain. Facial movements are variable, and their interpretation depends on body context, culture, situation, and deliberate expression [2, 5, 15]. Cardiac timing is also affected by respiration, posture, movement, fitness, medication, and sensor quality [16, 24, 28]. Neither channel should therefore be treated as a direct measurement of emotion.

Vibe Shuffle examines whether short-term changes within the same listener provide useful context for selecting the next song. At the beginning of a session, the system initializes personal reference values for facial and cardiac signals. During playback, the facial reference adapts slowly to longer visible changes, whereas the cardiac reference remains fixed. Later facial and cardiac observations are expressed as changes from the active references, while movement and nodding are summarized over a recent camera window. The resulting observations are mapped to relative valence and arousal coordinates. The observations collected during a song form a trajectory. Its mean identifies one of four regions in a fixed music catalog, and the nearest unplayed track in that region becomes the next recommendation. The region names Energetic, Calm, Tense, and Melancholic describe parts of the catalog rather than the listener’s emotional state. Physiological arousal and musical energy are likewise used as matching coordinates without being treated as the same construct.

Vibe Shuffle combines three elements:

- **Reference-informed sensing.** Reference-relative facial and cardiac observations are combined with recent-window movement cues without assigning a fixed emotion label.
- **Explicit next-song selection.** A trajectory mean, catalog region, and nearest-track rule turn those observations into one decision.
- **Exploratory comparison.** A within-session crossover across 15 sessions compared Vibe Shuffle with Random selection from the same catalog and measured momentary mood fit separately from general liking.

Mood-fit ratings were higher on average under Vibe Shuffle, whereas liking changed little. However, the uncertainty around both differences included zero, so the crossover did not establish improved recommendation quality. In a future music service, the method could re-rank tracks supplied by a conventional preference model. Evaluating that use will require validated signal mappings, complete trial-level records, and a preregistered study with a realistic candidate pool.

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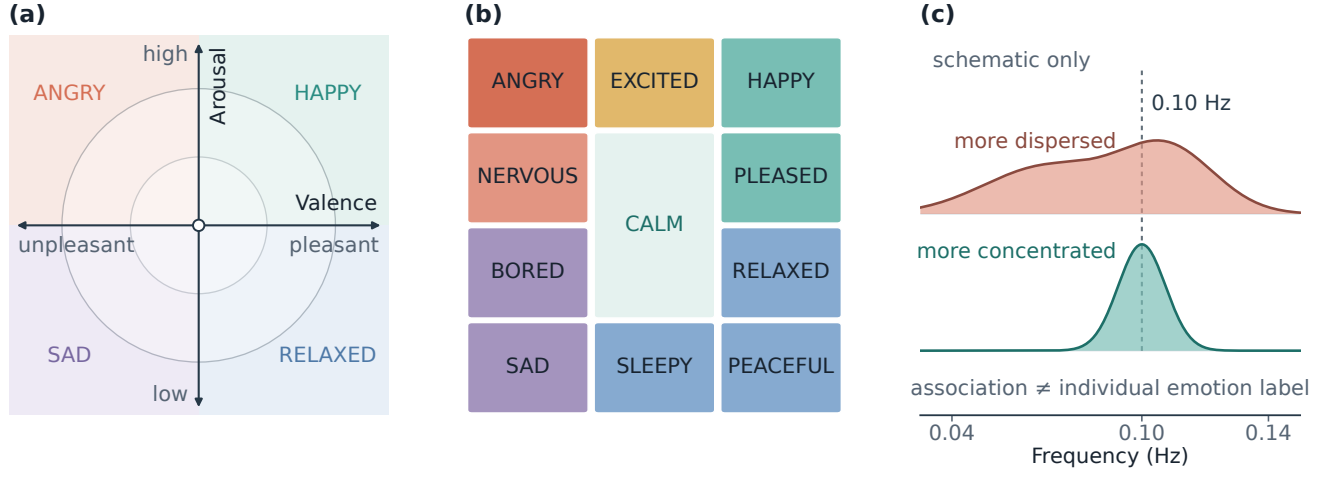


Figure 1: Panels (a)–(b) adapt the affect-space representations of Helmholtz et al. [10], based on Russell’s circumplex [25] and Thayer’s mood–arousal model [29]. Panel (c) schematically represents the group-level frequency distinction reported by Balaji et al. [4].

2 Background

Dimensional affect models provide a simple vocabulary for listeners and music. Russell’s circumplex places valence from unpleasant to pleasant and arousal from low to high activation [23, 25]. Thayer’s model adds categories such as energetic and tense [29]. These models are common in music-emotion research, but their meaning depends on the labels, stimuli, and context [7]. Vibe Shuffle uses them as a practical coordinate system for recommendation, not as a complete description of experience.

The continuous and categorical representations in Figure 1 define the recommendation space, while the HRV panel provides frequency context. Vibe Shuffle outputs neither emotion labels nor coherence profiles. Its coordinates combine changes from personal reference values with recent-window movement cues rather than representing direct measurements of emotion.

Songs are represented in a parallel valence–energy space. The catalog valence coordinate describes the positivity conveyed by a track, while energy captures its intensity and activity. Catalog valence and energy have served as matching axes for affective valence and arousal [10]. Musical energy and physiological arousal remain distinct constructs, and music-emotion coordinates are subjective and may change during a song [1, 31].

Splitting both axes at 0.5 creates four catalog regions. The mean coordinate of the listening trajectory determines the target region, and the remaining tracks in that region are ranked by their distance from this mean.

Cardiac timing provides one source of relative arousal evidence. HRV describes variation between heartbeats, while RMSSD summarizes short-term variation in RR intervals [28]. Their interpretation depends on recording length, respiration, movement, posture, fitness, medication, sensor quality, and device processing [16, 24]. Vibe Shuffle therefore uses heart rate and RMSSD only relative to a personal reference.

3 Related Work

Music can evoke, express, maintain, or regulate affect [12]. Emotion-aware recommenders therefore use the listener’s current context to guide song selection. *Feel the Moosic* asks listeners to choose a position in an affect space and maps it to Spotify valence and energy ranges [10]. Vibe Shuffle instead derives its coordinate from reference-relative facial and cardiac observations together with recent-window movement cues.

Other systems infer context from sensor data. Janssen et al. learned listener-specific relations between music and physiological responses and used them to guide an affective music player toward a target state [11]. Ayata et al. estimated valence and arousal from wearable measurements before recommending music [3]. Tran et al. matched facial estimates with nearby tracks in Spotify feature space [30]. Explicit reports, wearable measurements, and facial estimates can each provide recommendation context, but evidence that the resulting choices improve momentary fit remains limited.

Because facial and cardiac observations are uncertain and influenced by many factors, their role in song selection should remain understandable. Explanations can support transparency and trust [22], while perceived quality also depends on effort, privacy, and context [14].

Vibe Shuffle centers facial and cardiac observations on personal reference values, combines them with recent-window movement cues, aggregates the resulting coordinates as a trajectory, and applies a fixed region-and-distance rule.

4 Concept Overview

Vibe Shuffle expresses facial and cardiac observations as changes from personal reference values rather than population thresholds and combines them with recent-window movement cues. The resulting coordinates form a trajectory whose mean determines a

catalog region and selected track. Sensor outputs guide selection but do not identify the listener's emotion.

The selection objective is momentary track fit, not mood regulation. The method does not assume that an energetic track should reduce activation or that positive valence should counteract a negative state. Maintaining, intensifying, and shifting affect would require separate adaptation rules.

The method combines personal reference values, a multi-sample trajectory, a fixed region-and-distance rule, and a Random comparator using the same catalog. Missing sensing channels do not stop a session, and the active signal path is recorded.

Vibe Shuffle follows a six-step loop:

- (1) **Personal references.** Acquire facial and cardiac reference values.
- (2) **Observation.** Collect camera and cardiac observations during playback.
- (3) **Relative coordinates.** Combine reference-relative facial and cardiac observations with recent movement as valence-arousal points.
- (4) **Trajectory summary.** Summarize the collected points as a trajectory mean.
- (5) **Song selection.** Choose the nearest eligible song in the corresponding catalog region.
- (6) **Decision record.** Store the rating and decision fields.

Adaptation occurs between tracks, so brief fluctuations cannot interrupt the current song (Figure 2).

These design choices serve different purposes. Personal references make facial and cardiac evidence sensitive to changes within one listener rather than differences between people. Recent camera windows capture current movement. Updating only between tracks keeps playback stable. The four catalog regions provide an understandable target, while distance ranking selects the closest remaining song within that target. Optional sensing lets the same pipeline operate with camera data, cardiac data, both, or a fallback coordinate. The result is a direct and inspectable chain from observation to song choice.

5 Signal and Pipeline

The multimodal input has two distinct channels. A camera supplies visual and behavioral cues. An optional wearable supplies physiological timing. The facial reference runs for up to 14 seconds. The cardiac reference, when available, records 120 seconds of device-produced heart rate and RR intervals. The facial reference then adapts slowly to longer visible changes, whereas the cardiac reference remains fixed during the session. Subsequent facial and cardiac observations are centered on the active channel references. Movement and nodding are computed over a recent camera window rather than a personal movement reference. The references do not define a neutral or true emotional state. They provide local comparison points for the session.

Reference-relative facial observations primarily provide valence evidence, whereas cardiac changes provide relative activation evidence. Camera movement can additionally raise both coordinates. All mappings require independent validation.

Once the references are established, each channel contributes to the relative coordinate.

MediaPipe Face Landmarker runs locally and returns facial blend-shape activations [8, 18]. A fixed heuristic combines selected smile, cheek, brow, eye, jaw, frown, and mouth cues into positive, negative, and tension-related scores. Samples scheduled at 120 ms intervals are debiased against the reference and smoothed across frames. More positive visible evidence moves relative valence right. Negative or tension-related evidence moves it left. If no face is detected, valence returns to the center.

Visible movement and nodding can also add positive-valence and upward-arousal evidence. These cues are computed over a recent camera window and are not centered on a personal movement reference. The motion channel represents current observable behavior under a fixed rule, not felt emotion. Deliberate expression, dancing, and repositioning can change the visible behavior, while body context and culture constrain its interpretation [2, 5, 15]. Fixed coefficients and gating rules are applied uniformly across sessions.

Cardiac evidence enters through a compatible Bluetooth Heart Rate Service device, which supplies beats per minute and, when available, RR intervals. This interface exposes neither raw electrocardiography nor the device's beat-detection logic. Implausible or abruptly changing intervals are rejected before variability is calculated. The reference stores median heart rate, the root mean square of successive differences (RMSSD), and robust spread estimates.

During listening, higher-than-reference heart rate and lower-than-reference RMSSD provide upward arousal evidence. Changes in the opposite direction provide downward evidence. Heart rate receives the larger influence, and RMSSD is damped when the indicators disagree. At least 20 accepted RR intervals are required for the full estimate. RMSSD is a recognized short-term variability summary, but duration, respiration, posture, movement, fitness, medication, contact quality, and vendor processing affect its interpretation [16, 24, 28]. Agreement between 120-second and longer RMSSD measurements has been reported under controlled resting conditions, not short listening windows with movement [20].

SDNN and a custom spectral-peakedness measure termed physiology coherence are computed for diagnostic purposes only. This measure differs from the group-level coherence measure associated with post-session emotion labels in large-scale biofeedback data [4] and does not affect song selection.

The facial and cardiac values enter a shared pipeline for acquisition, plausibility filtering, centering on personal references, smoothing, and quality assessment. Camera-derived movement evidence is computed separately before coordinate fusion. Usable cardiac evidence sets the arousal base, while camera movement can add upward evidence. Camera movement supplies an upward-only fallback when cardiac data are unavailable. If neither channel is usable, the coordinate returns to the central reference point.

Camera frames, facial landmarks, and Bluetooth notifications are processed in browser memory. Stored study data comprise derived summaries, reference status, signal source, quality flags, timing, selected-song metadata, and ratings. Frames, landmarks, raw packets, waveforms, complete trajectories, personal facial-reference values, and the session seed are not stored.

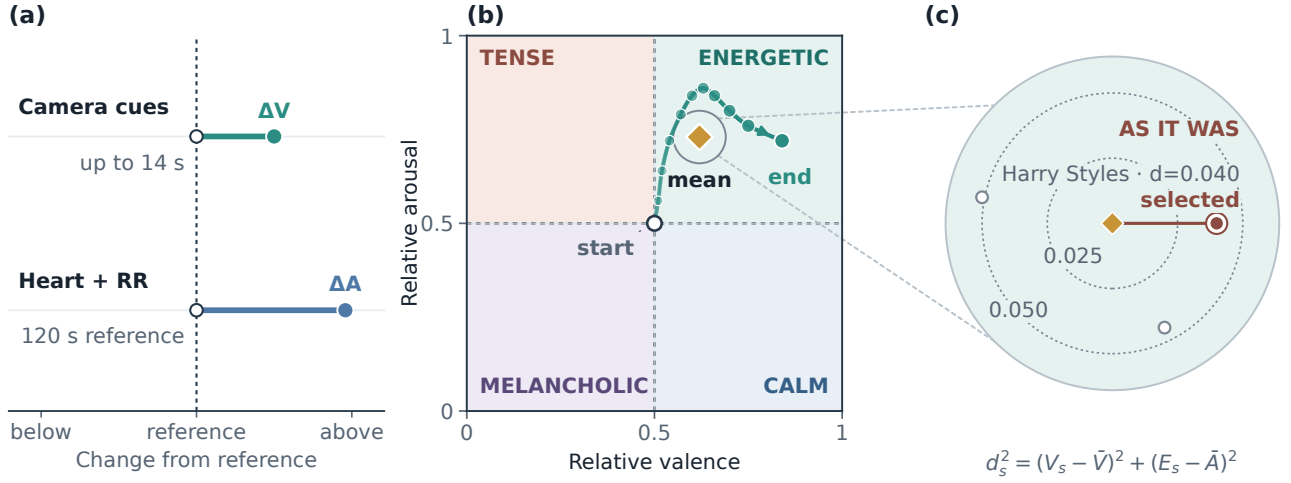


Figure 2: Panel (a) shows signed facial and cardiac changes measured against personal references. It shows the primary reference-relative signal paths. Camera movement may also contribute positive-valence and upward-arousal evidence. Panel (b) summarizes the trajectory between its open start and filled endpoint using the arithmetic mean (gold diamond). In panel (c), the nearest catalog track is *As It Was* by Harry Styles, which the selection rule therefore selects. Its Euclidean distance is shown beside the track. Two unlabeled open points indicate nearby alternatives, and the equivalent squared-distance expression appears below the panel. Panels (a)–(b) are schematic, whereas panel (c) uses catalog-derived coordinates and distance.

6 Inference and Adaptation

During confirmed playback, an event-driven update to the fused state appends a point to the trial trajectory

$$\mathcal{T} = \{(V_1, A_1), \dots, (V_m, A_m)\}.$$

At rating time, the arithmetic mean

$$(\bar{V}, \bar{A}) = \frac{1}{m} \sum_{i=1}^m (V_i, A_i)$$

becomes the target for the next Vibe choice. Aggregation reduces dependence on one observation but is event-weighted: periods that generate more fused-state updates contribute more points than periods with fewer updates. Sample count and listening duration are recorded for each trial.

Splitting both axes at 0.5 produces four catalog regions: Energetic, Calm, Tense, and Melancholic. These names identify valence-energy partitions. They do not classify the listener. Listener arousal and song energy are corresponding axes used for matching, not equivalent constructs [10, 31].

The frozen catalog contains 100 unique Spotify tracks, with 25 in each region. Track valence and energy coordinates were obtained from a public historical dataset [19].

Selection applies the same exclusion rule in both conditions: the current track and every previously played track are removed. Let $Q(\bar{V}, \bar{A})$ denote the set of remaining tracks in the catalog region containing the mean trajectory coordinate. Vibe Shuffle selects the smallest Euclidean distance:

$$s^* = \arg \min_{s \in Q(\bar{V}, \bar{A})} \sqrt{(V_s - \bar{V})^2 + (E_s - \bar{A})^2}.$$

Equal-distance candidates are ordered by a deterministic score derived from the track identifier and session seed. If the region is

exhausted, the method searches the complete remaining catalog. Both conditions use the same catalog, exclusion rules, number of tracks, playback interface, and rating questions. Vibe Shuffle selects the nearest remaining track in the target region, whereas Random ranks all remaining tracks by a deterministic pseudorandom score derived from the track identifier and a session-derived seed.

Because sensing is optional, a Vibe decision can use camera evidence, cardiac evidence, both, or neither. The first Random track is selected pseudorandomly, whereas a first-block Vibe track uses the fused coordinate available at session start. At a block transition, a first Vibe track uses the final listening window of the preceding block. Later Vibe tracks use the mean trajectory from the immediately preceding window.

Adaptation changes only the next-song ranking. At selection time, the method computes a target region, eligible set, distances, and selected track. Recorded selection variables include reference status, target coordinate, trajectory sample count, selected-track distance, region match, selected track, and quality indicators. Complete trajectories, eligible sets, and full distance lists are not stored.

7 User Study and Results

7.1 Study Design

The study examined whether context-sensitive selection improves the momentary fit of the next songs while maintaining general liking. Fifteen sessions followed a within-session crossover design. Each listener completed exactly one five-track Vibe block and one five-track Random block drawn from the same frozen catalog. The order was counterbalanced between listeners. One sequence presented Vibe before Random, while the other presented Random

before Vibe. No listener repeated the two-block sequence. Condition labels were hidden in the interface. Each session therefore provided one paired comparison between the two selection methods, while counterbalancing reduced systematic order effects across the sample.

Each track played for up to 60 seconds, with the option to continue to the rating screen earlier. After every track, listeners answered *How much do you like this song?* and *How well did it fit your current mood?* on seven-point scales. They also selected one of five descriptive mood categories: Energetic, Calm, Tense, Melancholic, or Neutral. This categorical response did not affect song selection and was not included in the analyses.

In the Vibe condition, song selection used the sensor-guided Vibe Shuffle mode deployed during the study. In the Random condition, selection did not use sensed context. Both conditions used the same catalog, number of tracks, playback interface, and rating questions. The comparison therefore evaluates the two delivered selection modes.

7.2 Analysis

The analysis compared the overall experience of the two five-track blocks. Ratings were averaged within each session and condition, and the resulting condition means served as the unit of analysis. Paired differences were defined as Vibe minus Random.

For mood fit and liking, Table 1 reports condition means and standard deviations, paired mean differences with two-sided 95% confidence intervals, and standardized paired effects d_z . Directional paired t tests and Wilcoxon signed-rank tests assess whether ratings were higher under Vibe. The confidence intervals use the paired t distribution with 14 degrees of freedom. Figure 3 additionally shows 20,000 fixed-seed bootstrap resamples of the paired mean distribution. Its displayed interval bars remain t -based. The analysis was exploratory and not preregistered, so estimation and two-sided uncertainty are primary [6, 21].

7.3 Results

Mood fit averaged 4.71 under Vibe Shuffle and 4.15 under Random selection. The paired difference was +0.56 rating points, with a 95% confidence interval from -0.21 to 1.33 and a standardized paired effect of $d_z = 0.40$. Eight session means were higher under Vibe, one was equal, and six were lower. The directional paired test yielded $t(14) = 1.56$, $p = .070$, and the signed-rank test yielded $W^+ = 74.0$, $p = .088$.

General liking averaged 4.56 under Vibe and 4.47 under Random. The paired difference was +0.09 points, with a 95% confidence interval from -0.52 to 0.71 and $d_z = 0.08$. The directional tests yielded $t(14) = 0.32$, $p = .375$, and $W^+ = 58.5$, $p = .353$. The mood-fit point estimate was larger than the liking estimate, but the two effects were not directly compared. Both confidence intervals include zero, so their size and direction remain uncertain.

8 Discussion, Limitations, and Future Work

8.1 Discussion

The two outcomes capture different aspects of recommendation quality. Liking reflects general preference, whereas mood fit asks whether a song suits the present situation. A context-aware method

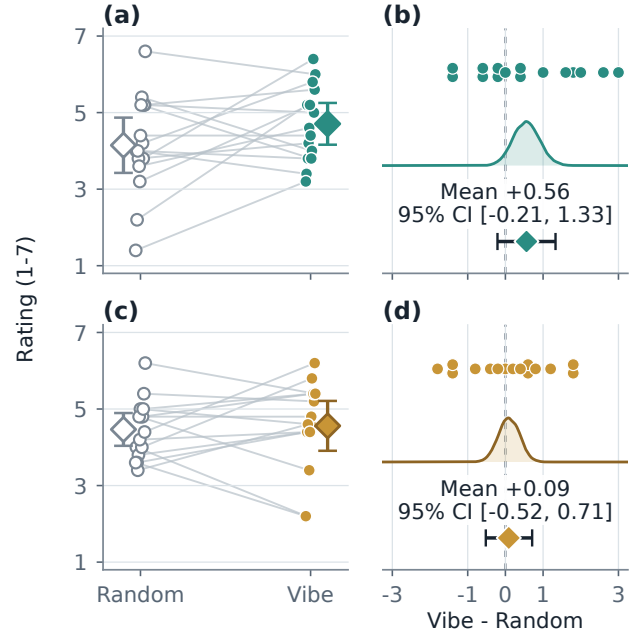


Figure 3: Mood fit is shown in panels (a)–(b) and liking in panels (c)–(d). The left panels connect the Random and Vibe five-track means for each session. Diamonds and bars summarize the condition means and their t -based 95% confidence intervals. The right panels show all paired differences, fixed-seed bootstrap distributions of the mean difference, and t -based 95% confidence intervals. Positive differences favor Vibe Shuffle.

can therefore target momentary fit without aiming to increase general liking. The outcomes were analyzed separately. Neither estimate was precise enough to support a performance conclusion.

The selection rule links reference-relative facial and cardiac changes together with recent-window movement cues to a recommendation through four explicit steps: personal references, a relative trajectory, a catalog region, and a distance ranking. This sequence keeps the recommendation logic understandable without assigning the listener an emotion label. Mood-fit differences nevertheless varied across sessions. The available averages cannot determine whether this variation arose from signal availability, trajectory shape, familiarity, order, or the songs remaining in the catalog.

Vibe Shuffle is intended to complement rather than replace preference modeling. A conventional recommender could first identify songs that the listener is likely to enjoy, after which the relative trajectory could re-rank that candidate set for the current session. Whether this combination improves momentary fit remains a hypothesis for future evaluation.

Vibe Shuffle also supports a lightweight deployment model. Camera frames, facial landmarks, and Bluetooth notifications are processed in browser memory, while stored data are limited to derived summaries, reference and quality status, timing, song metadata, and ratings. Raw camera and cardiac streams are therefore not

Table 1: Five-track condition means across 15 sessions. Differences are Vibe minus Random. Confidence intervals are two-sided. The p -values correspond to one-sided Vibe-over-Random tests.

Outcome	Vibe M (SD)	Random M (SD)	Difference [95% CI]	$t(14), p$	W^+, p	d_z
Mood fit	4.71 (0.99)	4.15 (1.30)	+0.56 [−0.21, 1.33]	1.56, .070	74.0, .088	0.40
Liking	4.56 (1.18)	4.47 (0.77)	+0.09 [−0.52, 0.71]	0.32, .375	58.5, .353	0.08

retained. Since camera and wearable inputs are optional, a session can continue with either channel, both channels, or a fallback coordinate. These properties make the method suitable as a re-ranking component in an existing music service.

Sixty-three deterministic unit tests cover catalog structure, data serialization, sensing heuristics, and song ranking. They check the implemented decision rules. They do not validate sensor accuracy or recommendation benefit.

8.2 Limitations

The fixed catalog and Random condition improve experimental control but do not reproduce candidate generation or preference modeling in a full music service. The two conditions could also select different songs and present them in a different order. The measured difference therefore reflects the complete recommendation experience produced by each method.

The crossover controls stable differences within a session, although practice, fatigue, and carryover may still affect the second block [9]. Condition labels were hidden in the interface, but study administration could still reveal them. Fifteen sessions leave broad confidence intervals and do not support stable order, subgroup, or hardware estimates. The small average liking difference does not establish that liking is preserved because no equivalence or non-inferiority range was defined. Five-track condition means can also hide differences between individual songs.

The sensing layer uses fixed mappings from facial behavior, movement, heart rate, and RMSSD to relative valence and arousal. These observations can change with visible behavior, body context, cultural interpretation, respiration, posture, physical movement, fitness, medication, and sensor quality [2, 5, 15, 16, 24, 28]. Movement may influence camera evidence and cardiac measurements simultaneously, and the study did not estimate the contribution of each sensing channel. The trajectory mean is event-weighted, because it is computed over event-driven fused-state updates rather than time-weighted samples. The four-region split also introduces hard boundaries at 0.5. These choices make the decision rule simple, but they are not the only ways to summarize and match a trajectory. The study did not separately measure sensor accuracy or playback failures, so their influence on the ratings is unknown.

8.3 Future Work

The sensing layer should be evaluated separately from song selection. Brief within-listener valence and arousal reports collected after selected trials could be compared with the system coordinates. Camera-only, cardiac-only, combined, and no-signal variants would then show which channels add useful information. These reports would assess the mapping rather than enter the recommendation rule.

The trajectory summary should also be studied directly. Event-weighted averaging can be compared with time-weighted averaging, a recent-window estimate, and robust summaries that reduce the influence of isolated changes. The region-first rule can be compared with continuous nearest-track matching in the full valence–energy space. These comparisons would separate the effects of sensing, temporal aggregation, and catalog matching.

A preference-based recommender provides the practical comparator. A standard model could generate a small candidate set of songs that the listener is likely to enjoy, and Vibe Shuffle could re-rank the same candidates using the current trajectory. Holding the candidate set fixed would isolate the value of session context from that of general preference modeling.

A larger preregistered crossover [17, 21] should use mood fit as the primary outcome and liking as a safeguard. Sample size should be based on the smallest mood-fit improvement that would matter in practice. A predefined equivalence or non-inferiority margin for liking would test whether contextual re-ranking improves momentary fit without making the selected songs less enjoyable. Track-level analyses could examine how effects develop across a session and differ between sensing variants. Effort, privacy, and willingness to enable a camera or wearable should be measured alongside recommendation quality [14].

9 Conclusion

Vibe Shuffle provides an explicit method for context-sensitive next-song selection. Reference-relative facial and cardiac observations are combined with recent-window movement cues in a short valence–arousal trajectory and translated into a catalog region and nearest eligible track. The system uses observable changes as recommendation context without presenting them as direct measurements of emotion.

Across 15 sessions, mood fit differed by +0.56 points and liking by +0.09 points, but both confidence intervals included zero. The available data therefore do not establish either effect. A larger study is needed to determine whether the method improves momentary fit while preserving general liking.

Vibe Shuffle links reference-relative facial and cardiac observations and recent movement cues to a next-song choice through a trajectory mean and a fixed region-and-distance rule. Validating the signal mappings and testing this rule as a re-ranking layer above a standard preference model are the next steps. This positions short-term, within-listener context as an inspectable complement to preference modeling, not a substitute for it.

10 Contribution Statement

Markus Baumann, Felix Hajuj, and Miriam Janner jointly developed the project concept, conducted the study, interpreted the results, and

reviewed the manuscript. Markus coordinated the implementation, analysis, figures, and preparation of the final paper.

Open Science and Transparency

To support transparency and reproducibility, the complete project repository is openly available at <https://github.com/eybmits/vibe-shuffle>. It includes the source code, catalog data, automated tests, figure-generation scripts, manuscript sources, rendered paper, and project website. Together, these materials allow others to inspect the system's decision logic, reproduce the software and paper outputs, and extend Vibe Shuffle in future research.

Large Language Model Assistance

GPT-5.6, Claude Fable 5, and Claude Opus 4.8 were used for debugging suggestions, grammar correction, and LaTeX formatting. The authors checked any retained suggestions against the code, data, literature, and generated outputs. They designed and implemented the system, conducted and analyzed the study, created the figures, wrote and revised the manuscript, and take full responsibility for the work.

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